

Amazon Alexa Product Reviews

Sentiment Analysis & Predictive Modelling Report

A Data Exploration and Machine Learning Study of 3,150 Verified Customer Reviews

Prepared using the amazon_alex dataset and accompanying data exploration & modelling notebook

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Table of Contents

Table of Contents.....	2
1. Executive Summary.....	1
2. Introduction.....	1
2.1 Business Context.....	2
2.2 Report Structure	2
3. Dataset Overview & Data Cleaning.....	2
3.1 Data Quality Checks	2
3.2 Data Types	3
4. Exploratory Data Analysis	3
4.1 Rating Distribution	3
4.2 Feedback Distribution	4
4.3 Product Variation Analysis	5
4.4 Review Length Analysis.....	7
5. Review Text & Word Cloud Analysis	10
5.1 Overall Vocabulary.....	10
5.2 Vocabulary Unique to Each Sentiment Class	10
6. Text Preprocessing & Feature Engineering.....	11
6.1 Bag-of-Words Vectorization	11
6.2 Train/Test Split & Scaling	11
7. Predictive Modelling	12
7.1 Random Forest Classifier	12
<i>Hyperparameter Tuning via Grid Search</i>	12
7.2 XGBoost Classifier	13
7.3 Decision Tree Classifier	13
8. Model Comparison & Evaluation	14
9. Conclusion & Recommendations	15
9.1 Key Takeaways.....	15
9.2 Recommendations for Future Work.....	15

1. Executive Summary

This report presents an end-to-end analysis of 3,150 verified customer reviews for Amazon Alexa-enabled devices, covering everything from the original Echo through more recent variants such as the Echo Dot, Echo Show, Echo Plus, Echo Spot, and Fire TV Stick configurations. The objective of the underlying study was twofold: first, to explore the structure and content of the review data to understand customer sentiment patterns, and second, to build and compare machine learning models capable of automatically classifying a review as positive or negative based on its text.

The dataset is heavily skewed toward positive sentiment: 91.87% of reviews are labelled positive (feedback = 1) and only 8.13% are labelled negative (feedback = 0). A star rating of 1 or 2 always corresponds to negative feedback, while a rating of 3, 4, or 5 always corresponds to positive feedback, confirming that the feedback label is a direct, deterministic function of the star rating rather than an independently derived sentiment score.

Three classification models were trained on a bag-of-words representation of the cleaned review text: Random Forest, XGBoost, and a Decision Tree. All three achieved strong test-set accuracy in the 92-94% range, with the Random Forest classifier performing best after hyperparameter tuning via grid search. Key findings and recommendations are summarized below:

- Customer sentiment is overwhelmingly positive across nearly every product variation, with Walnut Finish and Oak Finish variants recording the highest mean ratings (4.89 and 4.86 respectively).
- Negative reviews cluster around a distinct vocabulary — words such as "garbage", "pointless", "poor", "horrible", and "repair" — while positive reviews are dominated by words like "good", "amazing", "best", and "great".
- A tuned Random Forest classifier achieved 94.4% accuracy on the held-out test set, with a 10-fold cross-validation mean accuracy of 93.2%, making it the recommended model for production use.
- Because the dataset is imbalanced (roughly 11 positive reviews for every 1 negative review), all models are noticeably better at correctly identifying positive reviews than negative ones, which should be accounted for in any deployment decision.

2. Introduction

Understanding customer sentiment at scale is a central challenge for any consumer electronics business. Star ratings offer a coarse, numeric summary of customer satisfaction, but the free-text reviews that accompany them contain much richer information about what customers actually liked or disliked about a product. This report analyzes a corpus of Amazon Alexa product reviews with two goals in mind.

The first goal is exploratory: to characterize the dataset along its key dimensions — star rating, feedback label, product variation, and review length — and to surface the vocabulary that distinguishes positive

from negative reviews. The second goal is predictive: to build a supervised text classification pipeline that converts free-text reviews into a bag-of-words representation and trains machine learning models to predict whether a review is positive or negative, without needing to see the accompanying star rating.

2.1 Business Context

Automating sentiment classification at scale allows a business to triage incoming reviews, prioritize responses to dissatisfied customers, and track sentiment trends over time across different product variations without requiring a human to read every review individually. A reliable text-based classifier is also useful in settings where a star rating is unavailable but free-text feedback is present, such as social media mentions, support tickets, or survey responses.

2.2 Report Structure

Section 3 describes the dataset and the data-cleaning steps applied. Section 4 presents the exploratory data analysis of the rating, feedback, variation, and review-length fields. Section 5 examines the review text directly through word-frequency and word-cloud analysis. Section 6 describes the text preprocessing and feature engineering pipeline used to prepare the data for modelling. Section 7 details the three classification models that were trained, and Section 8 compares their performance. Section 9 concludes with key takeaways and recommendations for future work.

3. Dataset Overview & Data Cleaning

The dataset (amazon_alexas.tsv) is a tab-separated file containing 3,150 records and 5 original columns:

Column	Type	Description
rating	Integer (1-5)	Star rating given by the customer for the product variation
date	String	Date the review was submitted
variation	String	Product variation purchased (e.g. Black Dot, Charcoal Fabric)
verified_reviews	String	Free-text content of the customer's verified review
feedback	Integer (0/1)	Binary sentiment label: 0 = negative, 1 = positive

Table 1: Original dataset columns and descriptions

3.1 Data Quality Checks

A null-value check revealed exactly one record with a missing value in the verified_reviews column (a review with a rating of 2 and variation "White", dated 29-Jun-18). Since this single record cannot be used for text-based analysis or modelling, it was dropped from the dataset, reducing the total record count from 3,150 to 3,149. No other columns contained missing values.

A new engineered column, length, was added to the dataset, storing the character length of each review's verified_reviews text. This feature was later used to examine whether review length correlates with sentiment or rating.

3.2 Data Types

After cleaning, the dataset consists of three integer columns (rating, feedback, length) and two string columns (date, variation, verified_reviews — note verified_reviews and date and variation are all stored as string/object types). This confirms the data was loaded consistently with no type coercion issues.

4. Exploratory Data Analysis

4.1 Rating Distribution

The star ratings are heavily skewed toward the top of the scale. Of the 3,149 valid reviews, 2,286 (72.59%) received a 5-star rating, while only 161 (5.11%) received a 1-star rating.

Rating	Count	Percentage
5 stars	2,286	72.59%
4 stars	455	14.45%
1 star	161	5.11%
3 stars	152	4.83%
2 stars	95	3.02%

Table 2: Rating value counts and percentage distribution

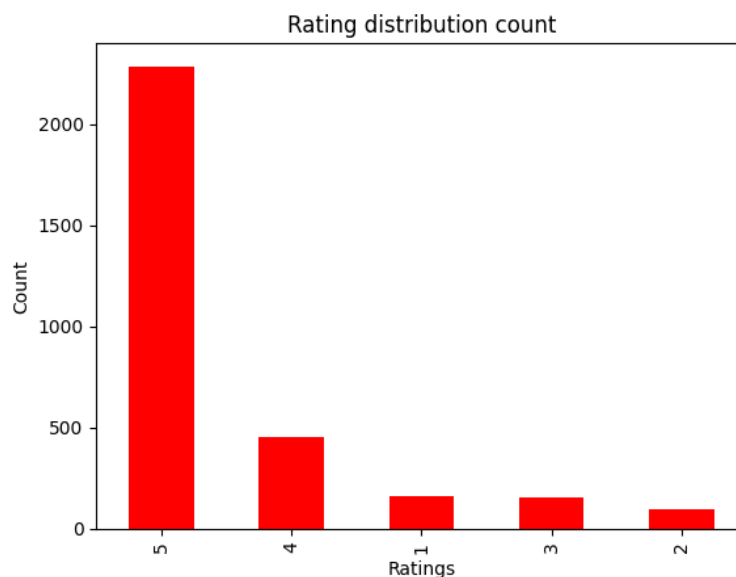


Figure 1: Bar chart of rating distribution counts

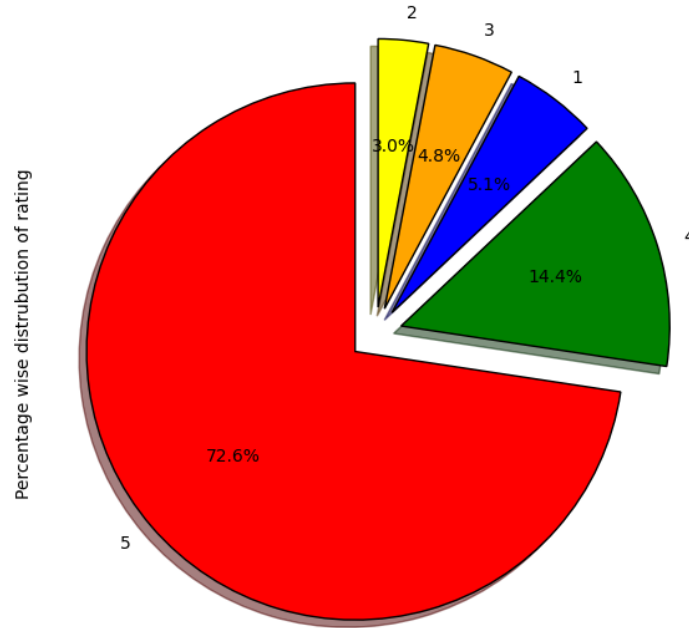


Figure 2: Pie chart of percentage-wise rating distribution

4.2 Feedback Distribution

The feedback column is binary, and the analysis confirms it is directly derived from the star rating: reviews rated 1 or 2 stars are always labelled feedback = 0 (negative), while reviews rated 3, 4, or 5 stars are always labelled feedback = 1 (positive). Overall, 2,893 reviews (91.87%) are positive and 256 reviews (8.13%) are negative — a substantial class imbalance that has implications for model evaluation discussed in Section 8.

Feedback	Count	Percentage
1 (Positive)	2,893	91.87%
0 (Negative)	256	8.13%

Table 3: Feedback value counts and percentage distribution

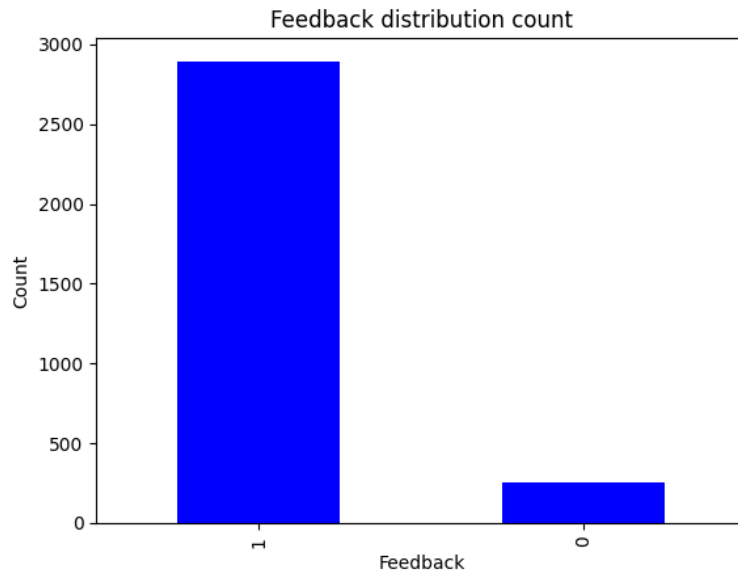


Figure 3: Bar chart of feedback distribution counts

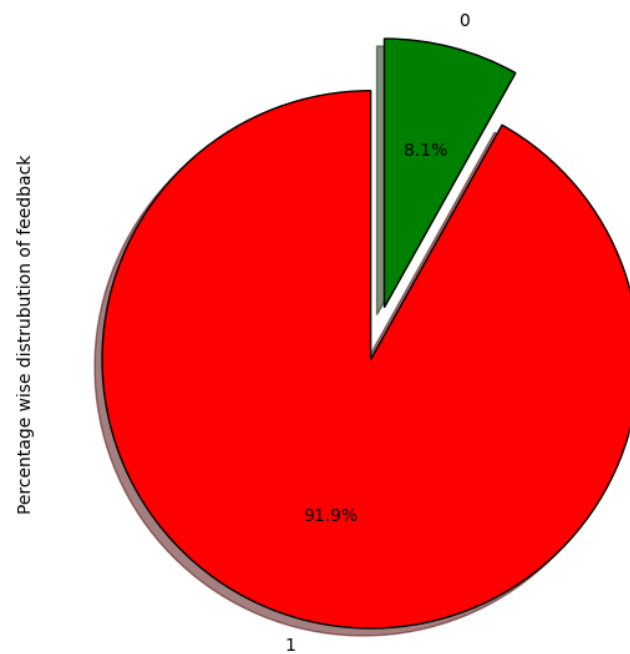


Figure 4: Pie chart of percentage-wise feedback distribution

Example of a negative (feedback = 0) review: "Sound is terrible if u want good music too get a bose."
 Example of a positive (feedback = 1) review: "Loved it!" These examples illustrate the clear qualitative difference in tone between the two classes.

4.3 Product Variation Analysis

The dataset spans 16 distinct product variations, with Black Dot (516 reviews) and Charcoal Fabric (430 reviews) being the most heavily represented, and Walnut Finish (9 reviews) and Oak Finish (14 reviews) the least represented.

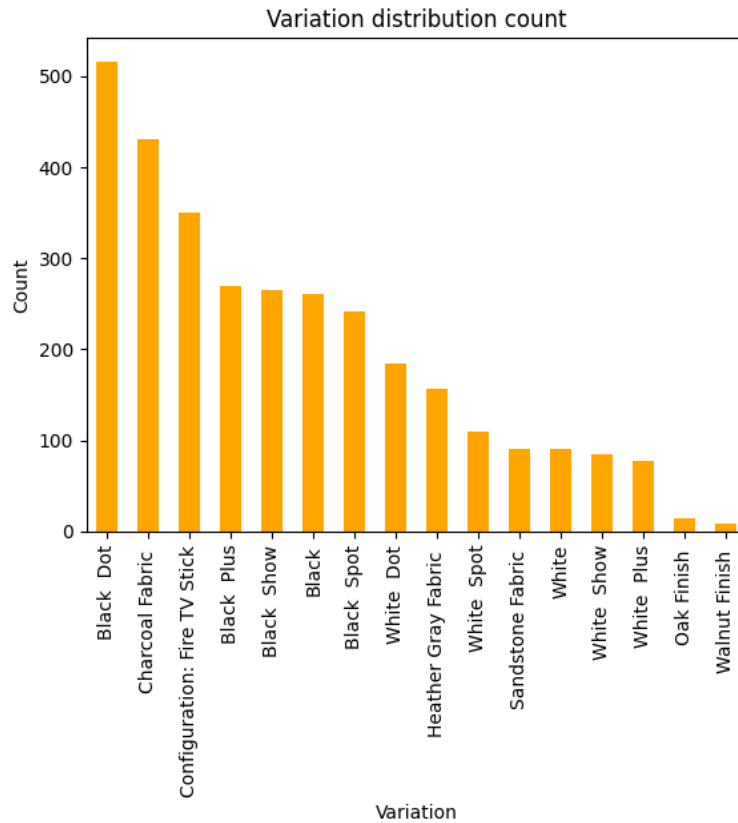


Figure 5: Bar chart of review counts by product variation

When mean rating is examined by variation, Walnut Finish (4.89), Oak Finish (4.86), and Charcoal Fabric (4.73) stand out as the most positively reviewed variations, while White (4.17) and Black (4.23) trend lowest — though still comfortably above the midpoint of the 1-5 scale.

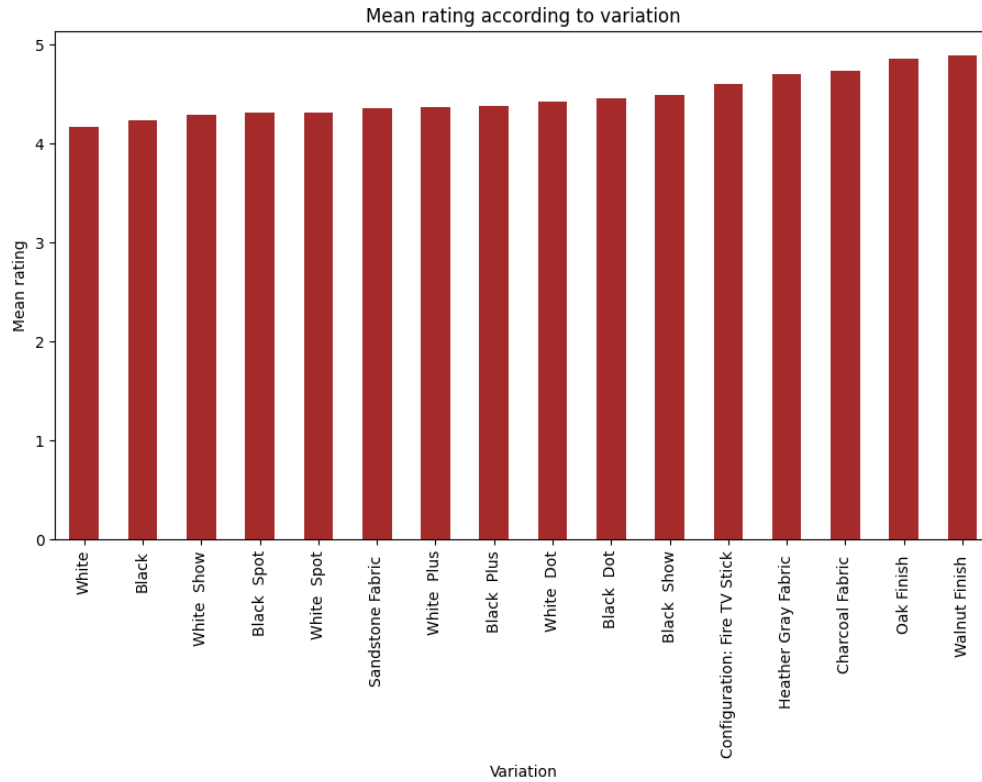


Figure 6: Mean rating by product variation, sorted ascending

4.4 Review Length Analysis

Review length (in characters) ranges from 1 to 2,853, with a mean of 132.7 characters, a median of 74 characters, and a standard deviation of 182.5 — indicating a right-skewed distribution where most reviews are short but a long tail of much longer, more detailed reviews exists.

Statistic	Value
Count	3,149
Mean	132.71
Std. Dev.	182.54
Min	1
25th percentile	30
Median (50%)	74
75th percentile	166
Max	2,853

Table 4: Descriptive statistics for review length (characters)

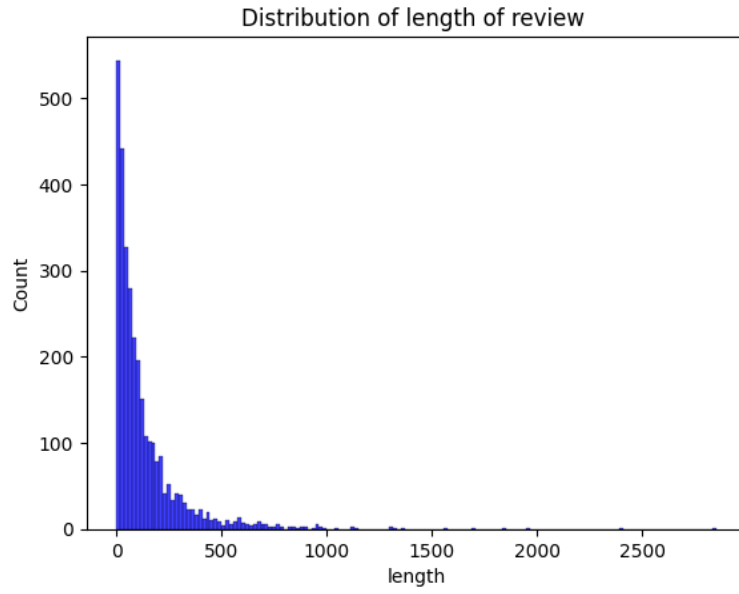


Figure 7: Distribution of review length, all reviews

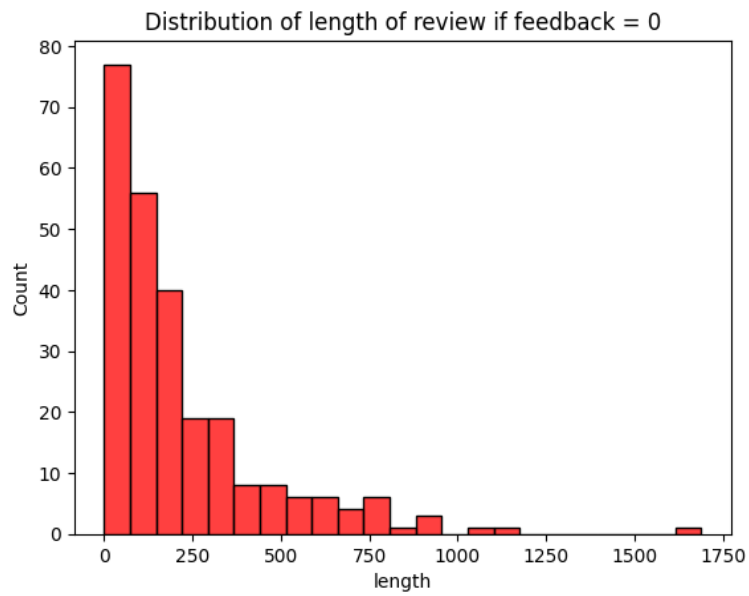


Figure 8: Distribution of review length, negative reviews (feedback = 0)

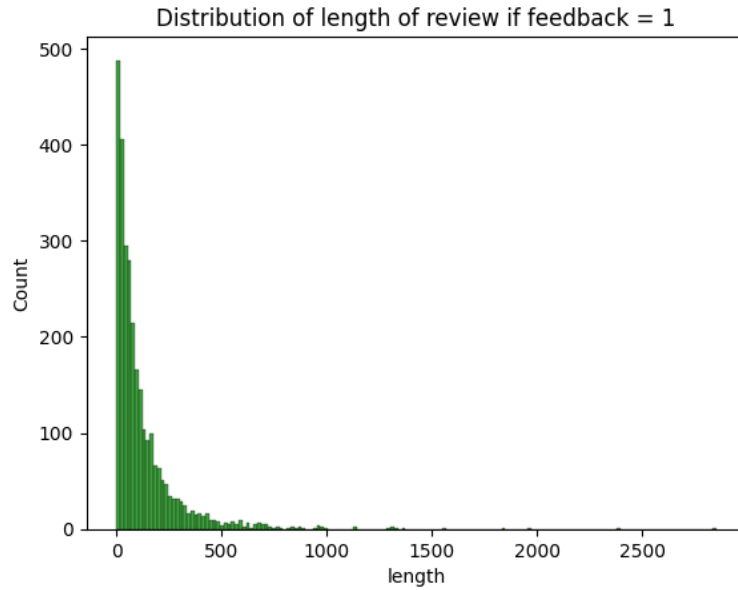


Figure 9: Distribution of review length, positive reviews (feedback = 1)

Both positive and negative reviews are dominated by short entries, but negative reviews show a somewhat heavier presence in the mid-length range, consistent with dissatisfied customers taking more time to explain what went wrong. The relationship between length and mean rating was also examined directly:

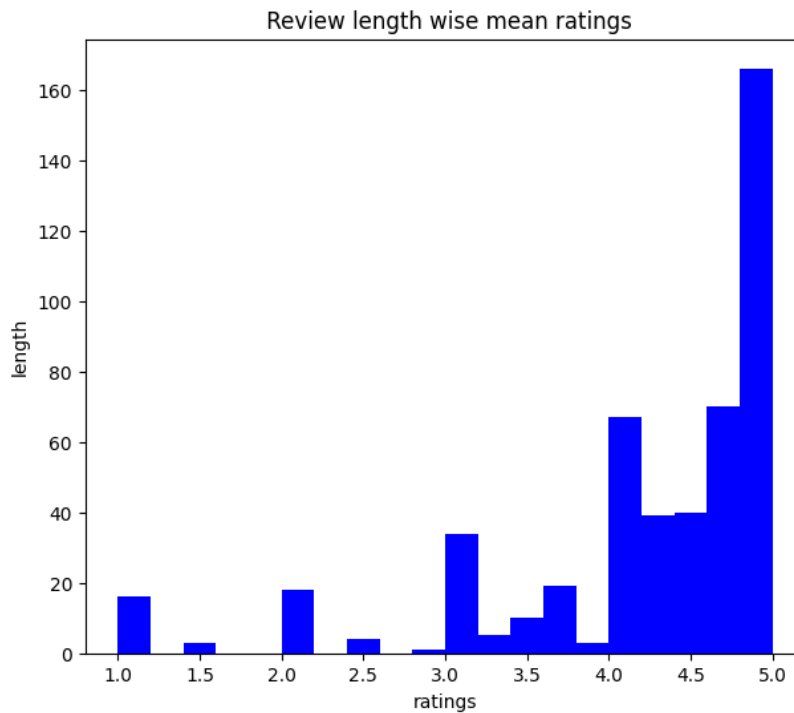


Figure 10: Histogram of mean rating grouped by review length

5. Review Text & Word Cloud Analysis

To understand the vocabulary driving sentiment, all review text was combined and visualized using word clouds, both overall and split by feedback class. A CountVectorizer (with English stop words removed) was first used to establish the vocabulary before building the word clouds.

5.1 Overall Vocabulary



Figure 11: Word cloud generated from all 3,149 reviews

The overall word cloud is dominated by product-related and generally positive terms such as "love", "great", "Echo", "music", and "works", which is expected given the 92% positive-review majority in the dataset.

5.2 Vocabulary Unique to Each Sentiment Class

To isolate the words that are genuinely distinctive of each sentiment, words appearing in negative reviews were filtered down to those that do not also appear in positive reviews, and vice versa for positive reviews.



Figure 12: Word cloud of vocabulary unique to negative reviews

The negative-only vocabulary surfaces clearly dissatisfied language, including terms such as "garbage", "pointless", "poor", "horrible", and "repair".



Figure 13: Word cloud of vocabulary unique to positive reviews

The positive-only vocabulary is dominated by enthusiastic and satisfied language, including terms such as "good", "enjoying", "amazing", "best", and "great". This clear lexical separation between the two classes is a strong signal that a bag-of-words classifier should be able to distinguish sentiment effectively, which is confirmed in Section 8.

6. Text Preprocessing & Feature Engineering

Before modelling, each review in `verified_reviews` was passed through a standard NLP cleaning pipeline to reduce noise and vocabulary sparsity:

- Non-alphabetic characters (punctuation, digits, symbols) were replaced with spaces using a regular expression.
- Text was converted to lowercase and split into individual word tokens.
- English stop words (e.g. "the", "is", "and") were removed using NLTK's stop word list.
- Remaining words were reduced to their root form using the Porter Stemmer (e.g. "loved" and "loving" both stem to "love").
- The cleaned, stemmed tokens for each review were rejoined into a single string, forming a cleaned text corpus of 3,149 documents.

6.1 Bag-of-Words Vectorization

The cleaned corpus was converted into a numerical feature matrix using scikit-learn's `CountVectorizer`, restricted to the 2,500 most frequent terms (`max_features = 2500`). This produced a feature matrix X of shape (3149, 2500) and a target vector y (the feedback column) of shape (3149,). The fitted vectorizer was serialized to disk (`countVectorizer.pkl`) so that it can later be reused to transform new, unseen reviews in the same way at inference time.

6.2 Train/Test Split & Scaling

The data was split into a 70% training set and a 30% test set (`random_state = 15` for reproducibility), yielding 2,204 training examples and 945 test examples, each with 2,500 features. Because the raw bag-of-words counts ranged up to a maximum value of 12 in the training set, a `MinMaxScaler` was fit on the

training data and applied to both the training and test sets, rescaling every feature to the [0, 1] range. This scaler was likewise saved (scaler.pkl) for consistent use at inference time.

Split	Rows	Columns
X (full feature matrix)	3,149	2,500
X_train	2,204	2,500
X_test	945	2,500

Table 5: Shapes of the feature matrix and train/test splits

7. Predictive Modelling

Three supervised classification algorithms were trained on the scaled bag-of-words features to predict the binary feedback label: a Random Forest classifier, an XGBoost classifier, and a Decision Tree classifier. Each model was evaluated using training and test accuracy, and confusion matrices were generated to inspect per-class performance.

7.1 Random Forest Classifier

A baseline Random Forest classifier (default scikit-learn hyperparameters) was first fit on the scaled training data, achieving a training accuracy of 99.46% and a test accuracy of 94.39% — a reasonably small gap suggesting the model was not drastically overfitting despite the very high training accuracy.

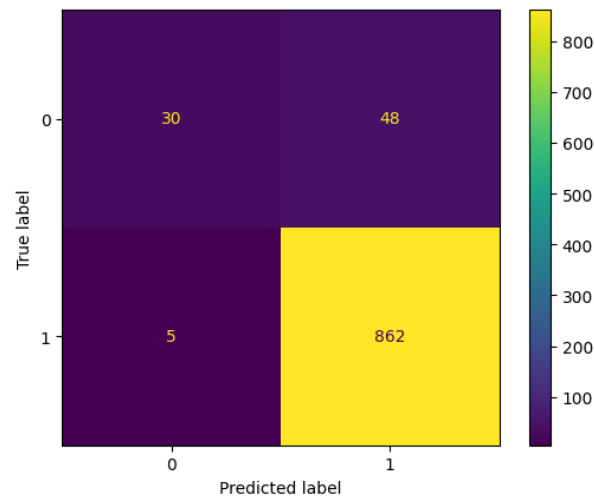


Figure 14: Confusion matrix for the baseline Random Forest classifier

10-fold cross-validation on the training set produced a mean accuracy of 93.19% with a very low standard deviation of 0.0076, indicating the model's performance is stable across different training subsets.

Hyperparameter Tuning via Grid Search

A grid search with stratified 2-fold cross-validation was run over combinations of bootstrap, max_depth (80 or 100), min_samples_split (8 or 12), and n_estimators (100 or 300). The best-performing combination found was bootstrap = True, max_depth = 100, min_samples_split = 8, and n_estimators = 100. Under this configuration, the mean cross-validated training accuracy was 96.80% and the mean cross-validated test accuracy was 92.20%, with a final held-out test accuracy of 94.39%.

7.2 XGBoost Classifier

An XGBoost classifier was trained on the same scaled features, achieving a training accuracy of 97.19% and a test accuracy of 93.76% — very close to, but marginally below, the Random Forest's test performance, with a noticeably smaller train/test gap.

	Predicted Negative	Predicted Positive
Actual Negative	35	43
Actual Positive	18	849

Table 6: Confusion matrix for the XGBoost classifier

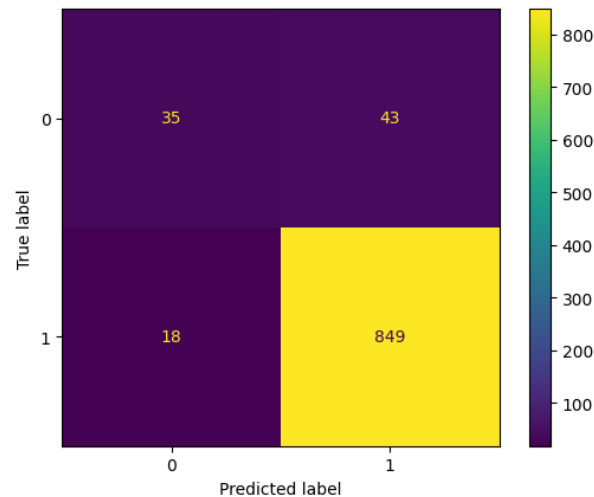


Figure 15: Confusion matrix plot for the XGBoost classifier

7.3 Decision Tree Classifier

A single Decision Tree classifier was trained as a simpler, more interpretable baseline. It achieved a training accuracy of 99.46% (identical to the unpruned Random Forest's training accuracy, as expected since Random Forests are ensembles of similarly unconstrained trees) but a lower test accuracy of 92.06%, the weakest of the three models on unseen data — consistent with a single decision tree's tendency to overfit relative to ensemble methods.

	Predicted Negative	Predicted Positive
Actual Negative	43	35

Actual Positive	95	772
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Table 7: Confusion matrix for the Decision Tree classifier

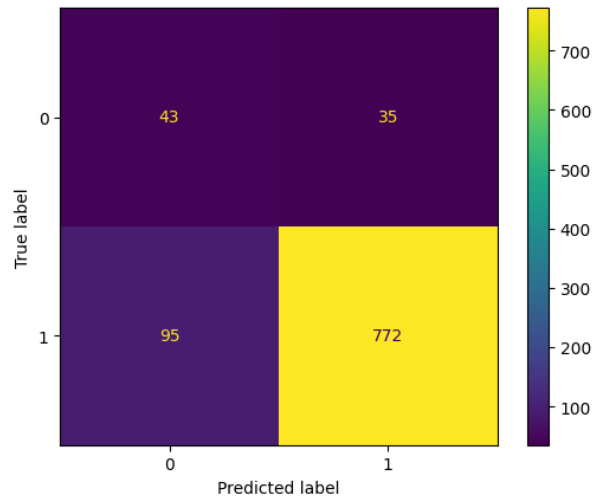


Figure 16: Confusion matrix plot for the Decision Tree classifier

8. Model Comparison & Evaluation

The table below summarizes training accuracy, test accuracy, and (where computed) cross-validated accuracy for all three models.

Model	Train Acc.	Test Acc.	10-fold CV Acc.
Random Forest (tuned)	99.46%	94.39%	93.19%
XGBoost	97.19%	93.76%	—
Decision Tree	99.46%	92.06%	—

Table 8: Comparison of model accuracy across the three classifiers

All three models perform comparably well, clustering within roughly two percentage points of each other on the held-out test set, with the Random Forest classifier achieving the highest test accuracy and the Decision Tree the lowest. Given that the dataset is imbalanced (about 92% positive vs. 8% negative), accuracy alone can be a somewhat optimistic metric: a naive classifier that always predicts "positive" would already score close to 92% accuracy. The confusion matrices in Section 7 show that all three models correctly identify the large majority of positive reviews (e.g. 849 of 892 actual positives correctly classified by XGBoost), but recall on the negative class is comparatively weaker (e.g. only 35 of 78 actual negatives correctly identified by XGBoost), reflecting the relative scarcity of negative training examples.

Based on test accuracy, cross-validation stability, and the confusion matrix breakdown, the tuned Random Forest classifier is recommended as the primary model, with XGBoost as a strong, faster-to-train

alternative. Both meaningfully outperform the single Decision Tree, confirming the value of ensemble methods for this task.

9. Conclusion & Recommendations

9.1 Key Takeaways

- Customer sentiment toward Amazon Alexa devices in this dataset is overwhelmingly positive (91.87%), with the feedback label deterministically derived from the star rating (1-2 stars = negative, 3-5 stars = positive).
- Premium finish variations (Walnut Finish, Oak Finish, Charcoal Fabric) received the highest average ratings, suggesting aesthetic/premium options correlate with higher satisfaction, though these variations also have far fewer reviews and should be interpreted with appropriate caution.
- Negative and positive reviews are lexically distinct, with clearly separable vocabularies that make bag-of-words text classification an effective approach for this task.
- A bag-of-words + Random Forest pipeline can classify review sentiment with approximately 94% test accuracy, and this pipeline (vectorizer, scaler, and trained model) has been serialized for reuse in a production setting.
- Model performance on the minority negative class lags behind performance on the majority positive class across all three algorithms, a direct consequence of class imbalance in the training data.

9.2 Recommendations for Future Work

- Apply class-imbalance techniques (e.g. class weighting, SMOTE oversampling, or under-sampling of the positive class) to improve recall on negative reviews specifically.
- Evaluate additional metrics beyond accuracy — precision, recall, F1-score, and ROC-AUC — per class, since accuracy alone can mask weak minority-class performance in an imbalanced dataset like this one.
- Experiment with richer text representations, such as TF-IDF weighting or pretrained word embeddings, in place of raw bag-of-words counts, which may improve the models' ability to separate subtle sentiment differences.
- Extend the analysis to identify sentiment trends over time (using the existing date column) and by product variation, to help product teams target quality or design improvements where they matter most.
- Consider a finer-grained multi-class target (predicting the 1-5 star rating directly, rather than the binary feedback label) to capture more nuance than a simple positive/negative split.